

Time Series

Applied Time Series Analysis **STA457 Lecture Notes**

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Properties of AR(1) - Second Proof

Using the lag operator L , the AR(1) process can be rewritten as follows:

$$AR(1) : (1 - \phi L)X_t = \delta + W_t. \quad (1)$$

Solving for X_t , we then get

$$X_t = \frac{\delta}{1 - \phi L} + \frac{1}{1 - \phi L} W_t, \quad (2)$$

which can formally be expanded to a geometric series, yielding

$$\begin{aligned} X_t &= (1 + \phi L + \phi^2 L^2 + \dots) \delta + (1 + \phi L + \phi^2 L^2 + \dots) W_t \\ &= (1 + \phi + \phi^2 + \dots) \delta + (W_t + \phi W_{t-1} + \phi^2 W_{t-2} + \dots). \end{aligned}$$

Since $|\phi| < 1$, we then obtain

$$X_t = \frac{\delta}{1 - \phi} + \sum_{j=0}^{\infty} \phi^j W_{t-j}. \quad (4)$$

Due to representation above, we get for the mean

$$\mathbb{E}[X_t] = \frac{\delta}{1 - \phi} + \sum_{j=0}^{\infty} \phi^j \mathbb{E}[W_{t-j}] = \frac{\delta}{1 - \phi}. \quad (5)$$

Properties of AR(1) - Second Proof (Cont'd)

For the variance we also get

$$\begin{aligned}
 \text{Var}[X_t] &= \mathbb{E}\left[\left(X_t - \frac{\delta}{1-\phi}\right)^2\right] = \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \phi^j W_{t-j}\right)^2\right] \\
 &= \mathbb{E}\left[(W_t + \phi W_{t-1} + \phi^2 W_{t-2} + \dots)^2\right] \\
 &= \mathbb{E}\left[W_t^2 + \phi^2 W_{t-1}^2 + \phi^4 W_{t-2}^2 + \dots + 2\phi W_t W_{t-1} + 2\phi^2 W_t W_{t-2} + \dots\right] \\
 &= \sigma_w^2(1 + \phi^2 + \phi^4 + \dots) = \frac{\sigma_w^2}{1 - \phi^2}.
 \end{aligned}$$

The autocovariance function can further be calculated as follows:

$$\begin{aligned}
 \text{Cov}(X_t, X_{t-h}) &= \mathbb{E}\left[\left(X_t - \frac{\delta}{1-\phi}\right)\left(X_{t-h} - \frac{\delta}{1-\phi}\right)\right] \\
 &= \mathbb{E}\left[\left(W_t + \phi W_{t-1} + \dots + \phi^h W_{t-h} + \dots\right)\left(W_{t-h} + \phi W_{t-h-1} + \phi^2 W_{t-h-2} + \dots\right)\right] \\
 &= \mathbb{E}\left[\left(W_t + \phi W_{t-1} + \dots + \phi^{h-1} W_{t-h+1} + \phi^h \left(W_{t-h} + \phi W_{t-h-1} + \phi^2 W_{t-h-2} + \dots\right)\right)\right. \\
 &\quad \left.\left(W_{t-h} + \phi W_{t-h-1} + \phi^2 W_{t-h-2} + \dots\right)\right] \\
 &= \phi^h \mathbb{E}\left[\left(W_{t-h} + \phi W_{t-h-1} + \phi^2 W_{t-h-2} + \dots\right)^2\right] \\
 &= \phi^h \text{Var}[X_{t-h}] = \frac{\phi^h \sigma_w^2}{1 - \phi^2}.
 \end{aligned}$$

Subsequently,

$$\rho_{t,t-h} = \text{Cov}(X_t, X_{t-h}) / \text{Var}[X_t] = \phi^h. \quad (8)$$

Properties of AR(1) - Second Proof (Cont'd)

Time Series Analysis

Random Walks

Random Walk With Drift

- 1 Random walk is a time series model where at each point of time the series moves randomly away from its current position.
- 2 The mathematical model for *random walk with drift* is obtained by replacing $\phi = 1$ in the AR(1) process, i.e.,

$$RW : X_t = \delta + X_{t-1} + W_t, \quad (9)$$

where $W_t \sim wn(0, \sigma_w^2)$.

- 3 Repeatedly substituting for past values gives

$$\begin{aligned} X_t &= \delta + X_{t-1} + W_t \\ &= 2\delta + X_{t-2} + W_{t-1} + W_t \\ &= 3\delta + X_{t-3} + W_{t-2} + W_{t-1} + W_t \\ &\vdots \\ &= \delta t + X_0 + \sum_{j=0}^{t-1} W_{t-j}. \end{aligned}$$

If the initial value, say X_0 , is fixed, then the mean value of random walk with drift is obtained as follows:

$$\mathbb{E}[X_t] = t\delta + X_0, \quad (11)$$

which depends on time, not just on lag.

Random Walk With Drift

- 1 The variance for a RW with drift is also obtained as:

$$\text{Var}[X_t] = \text{Var} \left[\delta t + X_0 + \sum_{j=0}^{t-1} W_{t-j} \right] = \text{Var} \left[\sum_{j=0}^{t-1} W_{t-j} \right] = \sum_{j=0}^{t-1} \text{Var} [W_{t-j}] = t\sigma_w^2. \quad (12)$$

- 2 The autocovariance function is further computed as being:

$$\begin{aligned} \text{Cov}(X_t, X_{t-h}) &= \text{Cov} \left(\sum_{i=0}^{t-1} W_{t-i}, \sum_{j=0}^{t-h-1} W_{t-h-j} \right) \\ &= \mathbb{E} \left[\left(\sum_{i=0}^{t-1} W_{t-i} \right) \left(\sum_{j=0}^{t-h-1} W_{t-h-j} \right) \right] \\ &= |t - h| \sigma_w^2. \end{aligned}$$

- 3 Both autocovariance and variance are functions of time, not the lag parameter alone.

Time Series Analysis

Data Generating Processes vs. Time Series Models

Strict Stationarity

❶ A stochastic process $\{X_t\}$ is said to be **strictly stationary** if for any $t_1, t_2, \dots, t_k$ and any lag parameter h , the joint distribution of $\{X_{t_1}, X_{t_2}, \dots, X_{t_k}\}$ is **identical** to that of $\{X_{t_1+h}, X_{t_2+h}, \dots, X_{t_k+h}\}$.

❷ A purely random process $\{W_t\} \stackrel{iid}{\sim} wn(0, \sigma_w^2)$ is strictly stationary since

$$\begin{aligned}\mathbb{P}[W_{t_1} \leq w_1, W_{t_2} \leq w_2, \dots, W_{t_n} \leq w_n] &= \mathbb{P}[W_{t_1} \leq w_1] \mathbb{P}[W_{t_2} \leq w_2] \dots \mathbb{P}[W_{t_n} \leq w_n] \\ &= \mathbb{P}[W_{t_1+h} \leq w_1] \mathbb{P}[W_{t_2+h} \leq w_2] \dots \mathbb{P}[W_{t_n+h} \leq w_n] \\ &= \mathbb{P}[W_{t_1+h} \leq w_1, W_{t_2+h} \leq w_2, \dots, W_{t_n+h} \leq w_n]\end{aligned}$$

❸ Strict stationarity requires, among other things, that for any t_1 and t_2 , the distributions of X_{t_1} and X_{t_2} must be the **same**. Strict stationarity further requires that **all bivariate distributions** of pairs (X_t, X_{t+h}) to be the **same** for all h .

❹ Note that with a strictly stationary process: $\mathbb{E}[X_t] = \mathbb{E}[X_{t+h}]$ for all t and h , so this implies that the mean function of any strictly stationary process is **constant** over time.

❺ Note that with a strictly stationary process: $\text{Var}[X_t] = \text{Var}[X_{t+h}]$ for all t and h , so this implies that the variance of any strictly stationary process is **constant** over time.

❻ For a strictly stationary process, we can express our autocovariance and autocorrelation coefficient functions simply in terms of the **time lag parameter** h , i.e.,

$$\gamma_h := \gamma_{t,t+h} \quad \text{and} \quad \rho_h := \rho_{t,t+h}. \quad (15)$$

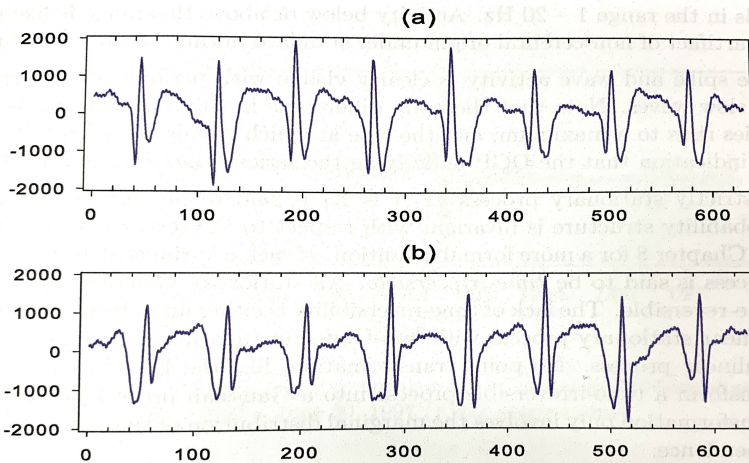


Figure 1: Plots of (a) original time series and (b) its reverse.

Weak Stationarity

- ❶ The requirements of strict stationarity are severe ones and usually difficult to establish mathematically. In fact, for most applications, the distributions involved are not even known. For this reason, less restrictive notions of stationarity have been developed, where the most common of these is **covariance stationary**.
- ❷ A stochastic process $\{X_t\}$ is said to be **weakly stationary** (or **covariance stationary**) if
 - ❶ $\mathbb{E}[X_t] = \mu$ (constant for all t)
 - ❷ $\text{Var}[X_t] = \sigma^2 < \infty$ (i.e., a finite constant for all t)
 - ❸ $\gamma_{t,s}$ depends only on the time lag $h = s - t$. In other words, we have $\gamma_h := \gamma_{t,t+h} = \gamma_{0,h}$ for every t and h .
- ❸ The strict stationarity implies the weak stationarity, however the vice versa is not necessarily true. In the remainder of the course, unless specified otherwise, the term **stationary** refers to **weak stationary**.
- ❹ In the special case that all joint distributions for the process are **multivariate normal**, then being **weakly stationary** is equivalent to being **strictly stationary** since the **moments** of the multivariate normal distribution completely specify the **laws** governing the stochastic process.

The following processes are weakly stationary.

AR(1) Process With $|\phi| < 1$

$$1 \quad \mathbb{E}[X_t] = \frac{\delta}{1 - \phi}$$

$$2 \quad \text{Var}[X_t] = \frac{\sigma_w^2}{1 - \phi^2}$$

$$3 \quad \gamma_{t,t+h} = \frac{\sigma_w^2 \phi^h}{1 - \phi^2}$$

MA(1) Process With $|\theta| < 1$

$$1 \quad \mathbb{E}[X_t] = \mu$$

$$2 \quad \text{Var}[X_t] = \sigma_w^2(1 + \theta^2)$$

$$3 \quad \rho_{t,t+h} = \theta/(1 + \theta^2) \text{ if } h = 1, \text{ and } \rho_{t,t+h} = 0 \text{ if } h \geq 2.$$

Random Cosine Wave $X_t = \cos[2\pi(t/\tau + \Phi)]$

$$1 \quad \mathbb{E}[X_t] = 0$$

$$2 \quad \text{Var}[X_t] = 1/2$$

$$3 \quad \gamma_{t,t+h} = \frac{1}{2} \cos[2\pi(\frac{h}{\tau})]$$

The random walk with drift is an example of nonstationary process.

RW With Drift

- ① $\mathbb{E}[X_t] = t\delta + X_0$
- ② $Var[X_t] = t\sigma_w^2$
- ③ $\gamma_{t,t+h} = |t - h|\sigma_w^2$

Difference Operator

- ① The *difference operator*, denoted by ∇ , is defined as follows:

$$\nabla X_t = X_t - X_{t-1} = (1 - L)X_t. \quad (16)$$

- ② The *ith difference operator* can be defined in terms of the lag operator as follows:

$$\nabla^i X_t = (1 - L)^i X_t. \quad (17)$$

- ③ Often applying the first difference operator ∇ to a nonstationary process, transforms it into a stationary one. For instance, let us apply ∇ to a random walk process with drift, which yields:

$$\nabla X_t = X_t - X_{t-1} = \delta + W_t, \quad (18)$$

which is a strictly stationary process.

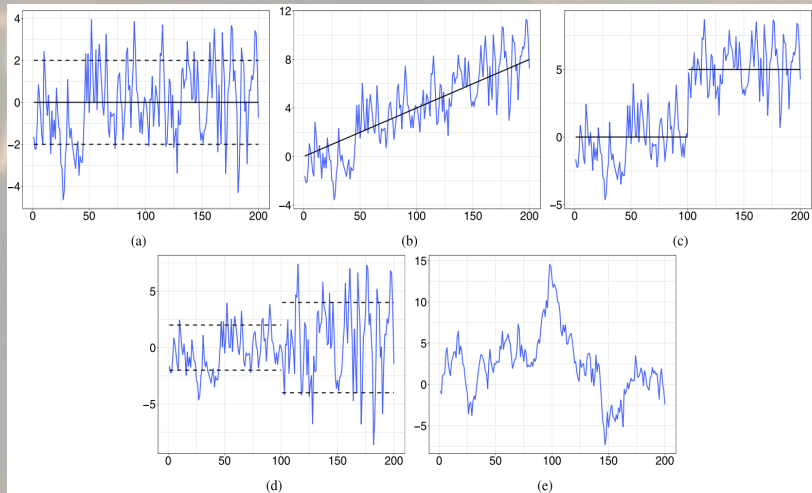


Figure 2: Examples of time series presenting the properties of (a) stationarity, and nonstationarity in the form of (b) trend stationarity, (c) level stationarity, (d) heteroscedasticity and (e) difference stationarity. The solid and dashed black lines represent the mean and the variance functions of the time series, respectively; courtesy to (Salles et al., 2019)

AR(1) Properties - Third Proof

Under the conditions of weak stationarity, the mean of X_t is constant. If we apply the expectation operator on

$$AR(1) : X_t = \delta + \phi X_{t-1} + W_t \quad (19)$$

we get:

$$\mathbb{E}[X_t] = \mathbb{E}[\delta + \phi X_{t-1} + W_t] = \delta + \phi \mathbb{E}[X_{t-1}] + \mathbb{E}[W_t]. \quad (20)$$

Because of $\mathbb{E}[W_t] = 0$ and $\mathbb{E}[X_t] = \mathbb{E}[X_{t-1}] = \mu$ for all t we can write

$$\mathbb{E}[X_t] = \mu = \frac{\delta}{1 - \phi}. \quad (21)$$

If we consider the deviations from the mean,

$$\tilde{X}_t = X_t - \mu \quad (22)$$

and substitute this in relation of AR(1), we get:

$$\tilde{X}_t + \mu = \delta + \phi(\tilde{X}_{t-1} + \mu) + W_t = \delta + \phi\tilde{X}_{t-1} + \phi\mu + W_t. \quad (23)$$

Then, it follows that

$$\tilde{X}_t = \delta + \mu(\phi - 1) + \phi\tilde{X}_{t-1} + W_t = \delta + \frac{\delta}{1 - \phi}(\phi - 1) + \phi\tilde{X}_{t-1} + W_t \quad (24)$$

So,

$$\tilde{X}_t = \phi\tilde{X}_{t-1} + W_t \quad (25)$$

AR(1) Properties - Third Proof (Cont'd)

If we multiply this equation by $\tilde{X}_{t-h}$ for $h > 0$ and take expectations from both sides, we get:

$$\mathbb{E}[\tilde{X}_{t-h}\tilde{X}_t] = \phi\mathbb{E}[\tilde{X}_{t-h}\tilde{X}_{t-1}] + \mathbb{E}[\tilde{X}_{t-h}W_t]. \quad (26)$$

Since $X_t = \delta/(1 - \phi) + \sum_{j=0}^{\infty} \phi^j W_{t-j}$, thus

$$\tilde{X}_{t-h} = W_{t-h} + \phi W_{t-h-1} + \phi^2 W_{t-h-2} + \dots, \quad (27)$$

yielding

$$\mathbb{E}[\tilde{X}_{t-h}W_t] = \begin{cases} \sigma_w^2 & \text{if } h = 0 \\ 0 & \text{if } h > 0 \end{cases} \quad (28)$$

Due to the stationarity assumption and considering the fact that autocovariance function is an even function, i.e., $\gamma_h = \gamma_{-h}$, eq. (26) results in

$$h = 0 : \mathbb{E}[\tilde{X}_t^2] = \phi\mathbb{E}[\tilde{X}_t\tilde{X}_{t-1}] + \sigma_w^2 \quad (29)$$

or

$$\gamma_0 = \phi\gamma_1 + \sigma_w^2 \quad (30)$$

Also,

$$h = 1 : \mathbb{E}[\tilde{X}_t\tilde{X}_{t-1}] = \phi\mathbb{E}[\tilde{X}_{t-1}^2] \quad (31)$$

or

$$\gamma_1 = \phi\gamma_0 \quad (32)$$

This leads to the variance of the AR(1) process

$$\gamma_0 = \frac{\sigma_w^2}{1 - \phi^2}. \quad (33)$$

AR(1) Properties - Third Proof (Cont'd)

Furthermore, for $h \geq 1$, eq. (26) implies

$$\gamma_1 = \phi\gamma_0 \quad (34)$$

$$\gamma_2 = \phi\gamma_1 = \phi^2\gamma_0 \quad (35)$$

$$\gamma_3 = \phi\gamma_2 = \phi^3\gamma_0 \quad (36)$$

$$\vdots \quad (37)$$

$$\gamma_h = \phi\gamma_{h-1} = \phi^h\gamma_0 \quad (38)$$

Thus, the covariances can be calculated using

$$\gamma_h - \phi\gamma_{h-1} = 0 \quad (39)$$

with the initial condition

$$\gamma_0 = \frac{\sigma_w^2}{1 - \phi^2}. \quad (40)$$

Now, the autocorrelation coefficient function is derived as being

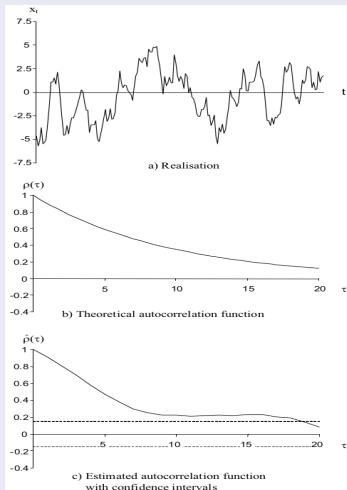
$$\rho_h = \frac{\gamma_h}{\gamma_0} = \phi^h, \quad h = 1, 2, 3, \dots \quad (41)$$

Time Series Analysis

Completing the Correlogram Discussion

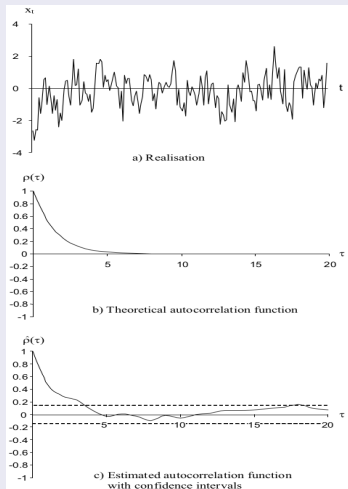
Correlogram as a Signature of Time Series

- Plots for $X_t = 0.9X_{t-1} + W_t$. The realization for $\phi = 0.9$ is relatively smooth. This is to be expected given the theoretical autocorrelation function because random variables with a considerable distance between each other still have high positive correlations.



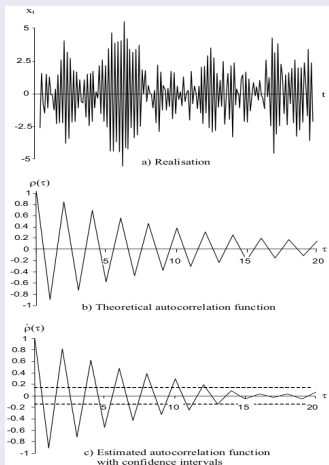
Correlogram as a Signature of Time Series

- Plots for $X_t = 0.5X_{t-1} + W_t$. The development of the realization with $\phi = 0.5$ is less systematic. The **geometric decrease** of the theoretical autocorrelation function is rather fast. The fourth order autocorrelation coefficient is only 0.0625.



Correlogram as a Signature of Time Series

- Plots for $X_t = -0.9X_{t-1} + W_t$. Contrary to previous cases, the realization of the AR(1) process with $\phi = -0.9$ follows a well-pronounced zigzag behaviour with, however, alternating positive and negative amplitudes. This is consistent with the theoretical autocorrelation function indicating that all random variables with even-numbered distance are positively correlated and those with odd-numbered distance are negatively correlated.



General Rules to Interpret Correlograms

- ➊ **AR(1) process:** It generally holds that the closer the parameter ϕ is to $+1$, the smoother the realizations will be. For negative values of ϕ we get zigzag developments which are the more pronounced the closer ϕ is to -1 . For $\phi = 0$ we get a pure random process (white noise).
- ➋ **A Random Series:** If a time series is completely random, then for large N , $r_k \approx 0$ for all non-zero values of k . For a completely random time series, 19 out of 20 of the values of r_k can be expected to lie between $\pm 1.96/\sqrt{N}$. However, plotting the first 20 values of r_k , one can expect to find one significant value on average even when time series is really random.
- ➌ **Short-term Correlation:** Stationary time series often exhibit short term correlation characterized by a fairly large value of r_1 followed by 2 or 3 significant coefficients (significantly greater than zero). The remaining coefficients tend to get successively smaller approaching zero. A time series which gives rise to such a correlogram is one for which an observation above the mean tends to be followed by one or more further observations above the mean and an observation below the mean tends to be followed by one or more further observations below the mean. An autoregressive model, may be appropriate for modeling series of this type.

General Rules to Interpret Correlograms (Cont'd)

- ➊ **Alternating Series:** If a time series has a tendency to alternate with successive observations on different sides of the overall mean, then the correlogram also tends to alternate. The value of r_1 will be negative, however, the value of r_2 will be positive as observation at lag 2 will tend to be on the same side of the mean.
- ➋ **Non-Stationary Series:** If a time series contains a trend, then the value of r_k will not come down to zero except for very large values of the lags. This is because a large number of observations lie on the same side of the mean since a trend exists. The sample autocorrelation function $\{r_k\}$ **should only** be calculated when the trend is removed from a nonstationary time series where the series is transformed to a stationary one.
- ➌ **Seasonal Fluctuations:** If a time series contains a seasonal fluctuation, then the correlogram will also exhibit an oscillation at the same frequency. If X_t follows a sinusoidal patterns then so does r_k .
- ➍ **General Remarks:** Experience is required to interpret autocorrelation coefficients. We need to study the probability theory of stationary series and different classes of models to be able to interpret the correlograms accurately.